Mathematical Analysis - Week 5

Lecture 16: Integrable and non-integrable functions

Two ways integrability can fail

Proposition 2.1.2. Riemann integrable on a bounded interval of positive length $\Rightarrow$ bounded.

Proof idea: if f is unbounded, some cell I_k of any finite partition has unbounded f. Fix all other tags; their sum is A. For any $T > 0$, choose c_k with

$$|f(c_k)|\Delta x_k > T + |A| \quad \Rightarrow \quad |S(f, P, \mathbf{c})| > T.$$

This works even for arbitrarily small mesh. No finite common limit.

Ex.2.1.2. $1/x$ and $1/x^2$ on $(0, 1)$: unbounded, hence not Riemann integrable. For $1/x$, just take $0 < c_1 < \min\{x_1, \Delta x_1/T\}$; the first term alone exceeds T.

Ex.2.1.3. $D = 1_{\mathbb{Q}}$ is bounded, but on $[a, b]$, $a < b$,

$$S(D, P, \mathbf{c}) = \begin{cases} b - a, & \text{all tags rational,} \\ 0, & \text{all tags irrational.} \end{cases}$$

Both choices work for every mesh. Boundedness alone is not enough.

Upper and lower sums

For bounded f on $[a, b]$, write $I_k = [x_{k-1}, x_k]$ and

$$M_k = \sup_{I_k} f, \qquad m_k = \inf_{I_k} f, \qquad \omega_k = M_k - m_k,$$
$$U(P) = \sum_k M_k \Delta x_k, \qquad L(P) = \sum_k m_k \Delta x_k,$$
$$\Delta(P) = U(P) - L(P) = \sum_k \omega_k \Delta x_k.$$

Always $L(P) \le S(f, P, \mathbf{c}) \le U(P)$. Use sup, inf; extrema need not be attained.

Refining P to Q gives $L(P) \le L(Q) \le U(Q) \le U(P)$. Define

$$\underline{J} = \sup_P L(P), \qquad \overline{J} = \inf_P U(P).$$

Theorems 2.1.3–5. For bounded f on $[a, b]$, the following are equivalent:

$$\begin{aligned} f \text{ integrable} &\iff \underline{J} = \overline{J} \\ &\iff \Delta(P) \to 0 \quad (\|P\| \to 0) \\ &\iff \forall \varepsilon > 0 \ \exists P : \ \Delta(P) < \varepsilon. \end{aligned}$$

Last criterion: only need to construct **one good partition** for each ε. Indeed $0 \le \overline{J} - \underline{J} \le U(P) - L(P) < \varepsilon$.

Corollary. For bounded f, non-integrability $\iff \exists \varepsilon_0 > 0\ \forall P : \Delta(P) \geq \varepsilon_0$.

Examples: $\Delta(D, P) = b - a$. For $f(x) = x^2$ on $[0, b]$,

$$\Delta(P) = \sum_k (x_k + x_{k-1})(\Delta x_k)^2 \leq 2b^2 \|P\| \to 0.$$

Uniform continuity

Definition 2.1.6. f is uniformly continuous on E if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in E : \quad |x - y| < \delta \ \Rightarrow\ |f(x) - f(y)| < \varepsilon.$$

Same δ everywhere. Ordinary continuity allows δ to depend on the base point.

To disprove it: find x_n, y_n with $|x_n - y_n| \to 0$ but a fixed positive output gap.

f on E	x_n	y_n	$\lvert f(x_n) - f(y_n)\rvert$
$1/x$ on $(0, 1]$	$1/n$	$1/(n+1)$	1
x^2 on $[1, \infty)$	n	$n + 1/n$	$2 + 1/n^2$
$\sin(1/x)$ on $(0, \pi)$	$1/(2\pi n)$	$1/(2\pi n + \pi/2)$	1

All three are continuous on their domains, none uniformly continuous.

Lemma 2.1.7. Continuous on $[a, b] \Rightarrow$ uniformly continuous.

Proof (subsequence trick). Otherwise, for some $\varepsilon_0 > 0$ choose

$$|x_n - y_n| < 1/n, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

Compactness gives $x_{n_j} \to c \in [a, b]$; then also $y_{n_j} \to c$. Continuity gives

$$|f(x_{n_j}) - f(y_{n_j})| \leq |f(x_{n_j}) - f(c)| + |f(y_{n_j}) - f(c)| \to 0.$$

Contradiction. Closedness keeps c in the domain; boundedness gives the subsequence.

From continuity to integrability

Theorem 2.1.8, continuous case. $f \in C[a, b] \Rightarrow f$ is Riemann integrable.

Proof: uniform continuity gives $\delta > 0$ such that

$$|x - y| < \delta \ \Rightarrow\ |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

If $\|P\| < \delta$, the maximum and minimum on each cell differ by less than $\varepsilon/(b - a)$. Hence

$$\Delta(P) = \sum_k \omega_k \Delta x_k < \frac{\varepsilon}{b - a} \sum_k \Delta x_k = \varepsilon.$$

The interval matters: continuous on $(0, 1]$ need not even be bounded $(1/x)$.

Small intervals around the bad points

Theorem 2.1.8, finite discontinuities. Bounded on $[a, b]$ + finitely many discontinuities $\Rightarrow$ integrable.

Proof / **Ex.2.1.8(a)**: let $|f| \le M$, $M > 0$. Cover the bad points by finitely many intervals of total length $\ell < \varepsilon/(4M)$; choose their other endpoints at continuity points. The remaining closed intervals have continuous f.

Partition the good pieces finely enough that their total oscillation sum is $< \varepsilon/2$. Include all covering endpoints in P. Then

$$\Delta(P) = \Delta_{\text{good}} + \Delta_{\text{bad}} < \frac{\varepsilon}{2} + 2M\ell < \varepsilon.$$

One good partition suffices. No need to control every fine partition again.

Ex.2.1.4. $f(x) = \sin(1/x)$ for $x > 0$, $f(0) = 0$: bounded, only discontinuous at 0, so integrable on $[0, 1]$. Fast oscillation near one point can be confined to a short interval.

Ex.2.1.5(a): restriction. If f is integrable on $[a, b]$, it is integrable on each $[u, v] \subseteq [a, b]$. Add u, v to a partition with small Δ; refinement decreases Δ, and the sum over $[u, v]$ is no larger than the whole sum.

Ex.2.1.5(b,c): finite changes. Changing finitely many values preserves integrability and the integral. For m changed points, let $K = \max |f - g|$ there. Each point belongs to at most two cells, so for the same tags,

$$|S(f, P, \mathbf{c}) - S(g, P, \mathbf{c})| \le 2mK\|P\| \to 0.$$

In particular, bounded functions have the same integrability on $[a, b]$ and (a, b) in the lecture's endpoint convention. Endpoint values do not affect the integral. Thus $\sin(1/x)$ is also integrable on $(0, 1]$.

Ex.2.1.6(a). Bounded f, continuous except possibly at $\{a_n\}$ with $a_n \to c$: still integrable.

Idea: one short interval around c contains the whole tail. Only finitely many exceptional points remain outside; cover those too. Make the total covered length $< \varepsilon/(4M)$ and use the same good/bad estimate.

Ex.2.1.6(b).

$$f(x) = \operatorname{sgn}\left(\sin \frac{\pi}{x}\right) \quad (0 < x \le 1), \qquad f(0) = 0.$$

Bounded by 1; continuous between consecutive $1/n$. Possible discontinuities: $1/n$ and 0, with only one accumulation point. Apply (a). Infinitely many discontinuities can still be allowed.

The Riemann function

Ex.2.1.7. Recall

$$\zeta(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms, } q > 0, \\ 0, & x \notin \mathbb{Q}. \end{cases} \qquad 0 \le \zeta \le 1.$$

Claim: $\displaystyle\int_a^b \zeta(x)\,dx = 0$ for every $a < b$.

Key observation: for any $\eta > 0$, only finitely many points in $[a, b]$ have $\zeta(x) \ge \eta$. Indeed $q \le 1/\eta$, and then only finitely many numerators p are possible.

Take $\eta = \varepsilon/[2(b-a)]$ and let N be the number of these points. In any partition, at most $2N$ cells contain one of them. On the other cells, $\sup \zeta \le \eta$. Thus

$$0 = L(P) \le U(P) \le 2N\|P\| + \eta(b-a) < \varepsilon \quad \text{if } \|P\| < \frac{\varepsilon}{4\max\{N, 1\}}.$$

$L(P) = 0$ because every cell contains irrationals. All tagged sums are squeezed to 0.

Ex.2.1.8(b): one-partition version. Cover the same finite set by intervals of total length $< \varepsilon/2$, with irrational interior covering endpoints. Use these endpoints to form P. On the covered cells use $\omega \le 1$; on the rest use $\omega \le \eta$. Then

$$\Delta(P) < 1 \cdot \frac{\varepsilon}{2} + \eta(b-a) = \varepsilon.$$

The important set here is where the values are **large**, rather than all rational points.

What to estimate

Continuous on $[a, b]$	uniformly small cell oscillation
Finitely many bad points / one convergent sequence	small total length around bad points
Riemann function ζ	finitely many large values

D versus ζ: both are nonzero on the rationals. For D, every cell has oscillation 1; for ζ, the large spikes can be covered cheaply. The quantity to control is always

$$\boxed{\Delta(P) = \sum_k (\sup_{I_k} f - \inf_{I_k} f)\,\Delta x_k.}$$